Solution

1. (Binary stones). Consider the following game: two players may take from a heap any number of stones that is a power of 2 (including $1 = 2^0$). The player who has taken the last stone wins. For which number of stones does the first player win? How should he play? (You should not only specify the strategy but also prove that it works.)
Solution.
Let n be the number of stones. Allowed moves are to remove $1, 2, 4, 8, \ldots, 2^k$ stones. We seek the losing positions:

Small cases.

n	0	1	2	3	4	5	6	7	8	9
outcome	lose	win	win	lose	win	win	lose	win	win	lose

The losing positions among these are $0, 3, 6, 9, \ldots$

Claim. The losing positions are exactly the multiples of 3. Equivalently, the first player wins if and only if $n \not\equiv 0 \pmod 3$. For example, 1, 2 or 4 and etc.

Proof. Powers of 2 modulo 3 alternate:

$$2^0 \equiv 1 \pmod 3 \qquad 2^1 \equiv 2 \pmod 3$$

$$2^2 \equiv 1 \pmod 3 \qquad 2^3 \equiv 2 \pmod 3$$

In general $2^k \equiv (-1)^k \pmod 3$, so every allowed move removes either 1 or 2 modulo 3, never 0 modulo 3.
If $n \equiv 0 \pmod 3$, then for any move $n \mapsto n - 2^k$ we have $n - 2^k \not\equiv 0 \pmod 3$.
If $n \not\equiv 0 \pmod 3$, there is a move to a multiple of 3:

1. $n \equiv 1 \pmod 3$, remove $1 = 2^0$
2. $n \equiv 2 \pmod 3$, remove $2 = 2^1$

Thus from a multiple of 3 every move goes to a non-multiple of 3, and from a non-multiple of 3 there is a move to a multiple of 3. The terminal position $n = 0$ is a multiple of 3 and is losing for the player to move. Hence multiples of 3. Therefore the first player wins if and only if $n \not\equiv 0 \pmod 3$.

Optimal strategy. If $n \equiv 1 \pmod 3$, remove 1 stone; if $n \equiv 2 \pmod 3$, remove 2 stones. Thereafter always leave a heap size divisible by 3. Since the opponent can only subtract 1 or 2 modulo 3 from the heap, you can always restore divisibility by 3 on your turn. Eventually the opponent faces $n = 0$ and loses.

2. (Moving the knight). A knight is standing at the top-right corner of the chessboard (field `h8`). Two players move it sequentially, but they must move it two fields downwards or leftwards (and then one field in any suitable direction). For instance, from `f6` the knight can be moved only to `g4`, `e4`, `d5` or `d7`. The player unable to make a move loses. Which player does have a winning strategy? What is this strategy? Can you generalize the answer to an arbitrary size of the field?

Solution.

We number the board by coordinates

$$a1 = (1, 1), \qquad h8 = (8, 8).$$

The knight starts at

$$h8 = (8, 8).$$

A legal move is one of

$$\begin{aligned}(x, y) &\mapsto (x + 1, y - 2),\\ (x, y) &\mapsto (x - 1, y - 2),\\ (x, y) &\mapsto (x - 2, y + 1),\\ (x, y) &\mapsto (x - 2, y - 1)\end{aligned}$$

provided the new square is still on the board. winning if the player to move from it can move to a losing position.

Losing positions on the 8×8 board.

In coordinates, these are the 2×2 blocks

$$\begin{gathered}\{1, 2\} \times \{1, 2\},\\ \{1, 2\} \times \{5, 6\},\\ \{5, 6\} \times \{1, 2\},\\ \{5, 6\} \times \{5, 6\},\end{gathered}$$

together with the special square

$$(8, 8).$$

We can see that losing positions satisfy

$$x, y \equiv 1 \text{ or } 2 \pmod 4$$

On the 8×8 board, the losing positions can be displayed as follows:

	1	2	3	4	5	6	7	8
8	.	.	.	.	.	.	.	L
7	.	.	.	.	.	.	.	.
6	L	L	.	.	L	L	.	.
5	L	L	.	.	L	L	.	.
4	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.
2	L	L	.	.	L	L	.	.
1	L	L	.	.	L	L	.	.

Why these positions are losing.
Let
$$R = \{(x, y) : x, y \equiv 1 \text{ or } 2 \pmod 4\}.$$
These positions form repeated 2×2 blocks. A knight move changes one coordinate by 1 and the other coordinate by 2. Therefore a move from a position in R always leaves R. So no position in R has a move to another position in R.
Conversely, by backward induction, every legal move out of one of these blocks leads to a position from which the next player can move back into a smaller R-position. Hence the positions in R are losing.
Now check the starting square:
$$h8 = (8, 8)$$
Its only legal moves are
$$(8, 8) \mapsto (7, 6)$$
and
$$(8, 8) \mapsto (6, 7)$$
Both of these positions are winning, because both can move to
$$(5, 5) = e5,$$
which is losing:
$$(7, 6) \mapsto (5, 5),$$
$$(6, 7) \mapsto (5, 5)$$
Therefore every legal move from $h8$ goes to a winning position. Hence
$$h8$$
is losing.
Thus the first player loses and the second player wins.

General solution.
Consider an $n \times n$ board with coordinates
$$(1, 1), (1, 2), \ldots, (n, n),$$
where the knight starts at the top-right corner
$$(n, n).$$
The legal moves are the same as before whenever the new square is still on the board.

Losing blocks.
The basic losing positions are the 2×2 blocks satisfying
$$x, y \equiv 1 \text{ or } 2 \pmod 4.$$

A move from such a square always leaves this set. Moreover, after any legal move from such a square, the next player can move back to a smaller square of the same type. Therefore these positions are losing positions.

Starting from the corner.
Now consider the starting square

$$(n, n).$$

There are four cases.

Case 1: $n \equiv 1 \pmod 4$.
Then

$$(n, n) \equiv (1, 1) \pmod 4,$$

so (n, n) lies in a losing block. Hence the first player loses.

Case 2: $n \equiv 2 \pmod 4$.
Then

$$(n, n) \equiv (2, 2) \pmod 4,$$

so (n, n) also lies in a losing block. Hence the first player loses.

Case 3: $n \equiv 3 \pmod 4$.
Then the first player can move

$$(n, n) \mapsto (n - 1, n - 2)$$

Modulo 4, this is

$$(3, 3) \mapsto (2, 1)$$

Since $(2, 1)$ is a losing-type residue, the first player can move to a losing position. Hence the first player wins.

Case 4: $n \equiv 0 \pmod 4$.
The square (n, n) is not itself in the losing block pattern. However, because it is the top-right corner, it has only two legal moves:

$$(n, n) \mapsto (n - 1, n - 2),$$

$$(n, n) \mapsto (n - 2, n - 1)$$

Both resulting positions are winning, because from either one the next player can move to

$$(n - 3, n - 3).$$

Indeed,

$$(n - 1, n - 2) \mapsto (n - 3, n - 3),$$

and

$$(n - 2, n - 1) \mapsto (n - 3, n - 3)$$

Since

$$n \equiv 0 \pmod 4,$$

we have

$$n - 3 \equiv 1 \pmod 4$$

Therefore

$$(n-3, n-3) \equiv (1,1) \pmod 4,$$

which is a losing block position.
Thus every legal move from (n, n) gives the opponent a move to a losing position. Hence (n, n) is losing.

Conclusion.
Therefore, on an $n \times n$ board starting from the top-right corner,

$$\text{the first player wins if and only if } n \equiv 3 \pmod 4$$

For the usual chessboard,

$$8 \equiv 0 \pmod 4,$$

so the first player loses and the second player wins.

3. (Pawn duel). Consider the following game on a 3×8 table. Each of the two players has three pawns along the short edges. During the first turn each player may move any of his pawns towards the opposite edge but not further than by 3 cells. During the later rounds each player may move any of his pawns towards the opposite edge by any number of cells. The players cannot move the pawns vertically or diagonally, nor can they jump across or take pawns of the other player. The player that has no legal turn loses. Which player has a winning strategy? How should he play?
Solution.
For each of the three rows, let

$$d_i = \text{the number of empty cells between the two pawns in row } i$$

Initially, in each row the two pawns stand at opposite ends of an 8-cell row, so

$$d_1 = d_2 = d_3 = 6$$

Thus the initial position is

$$(6, 6, 6)$$

A move in one row simply decreases exactly one of the numbers d_i by a positive amount. Hence, after the first restricted moves, the game is equivalent to Nim with three heaps of sizes

$$d_1, d_2, d_3$$

In Nim, a position is losing if and only if

$$d_1 \oplus d_2 \oplus d_3 = 0,$$

where $\oplus$ denotes bitwise XOR.
The initial Nim-sum is

$$6 \oplus 6 \oplus 6 = 6 \neq 0$$

Indeed, in binary this is

$$110_2 \oplus 110_2 \oplus 110_2 = 110_2$$

The first player should move one pawn forward by exactly 2 cells. This changes one gap from 6 to 4, so the position becomes

$$(4, 6, 6).$$

Now

$$4 \oplus 6 \oplus 6 = 4$$

If the second player wanted to make the Nim sum zero immediately, he would have to change one heap h into

$$h' = h \oplus 4$$

The possible changes for h are :
$h = 4$:

$$4 \to 4 \oplus 4 = 0,$$

or for $h = 6$:

$$6 \to 6 \oplus 4 = 2$$

Both moves require decreasing a gap by 4. But on the second player's first move, he is allowed to move a pawn by at most 3 cells. Therefore, the second player cannot move to a zero Nim-sum position.
After that, the first player is no longer restricted. From then on he uses the standard Nim strategy: after every move of the opponent, make

$$d_1 \oplus d_2 \oplus d_3 = 0$$

For example, after the first move

$$(6, 6, 6) \to (4, 6, 6),$$

the second player's first move can produce, up to permutation of rows, only one of the following positions:

$$(3, 6, 6), \quad (2, 6, 6), \quad (1, 6, 6),$$

or

$$(4, 5, 6), \quad (4, 4, 6), \quad (4, 3, 6)$$

The first player answers as follows:

$$(3, 6, 6) \to (0, 6, 6),$$

$$(2,6,6) \to (0,6,6),$$
$$(1,6,6) \to (0,6,6),$$
$$(4,5,6) \to (3,5,6),$$
$$(4,4,6) \to (4,4,0),$$
$$(4,3,6) \to (4,2,6)$$

In each case the resulting Nim-sum is zero.
Thus the first player can always give the second player a position satisfying

$$d_1 \oplus d_2 \oplus d_3 = 0.$$

The game eventually ends. Since the first player always leaves the second player a zero Nim sum position, the second player is the one who eventually has no legal move.

Therefore, the first player has a winning strategy.

The first player wins.

His strategy is:

First move: move any pawn forward by exactly 2 cells.

After that, always move so that

$$d_1 \oplus d_2 \oplus d_3 = 0$$

4. (Cake division).
Albert and Bernie divide a cake by alternating proposals. First Albert proposes a division. If Bernie accepts, the division is implemented. Otherwise they wait one hour and Bernie proposes. If Albert accepts, the division is implemented; otherwise they wait another hour, and so on.
A fraction p of the cake is worth

$$p\left(\frac{1}{2}\right)^t$$

after t hours of waiting, for $t \leq 5$. After 5 hours the cake becomes worthless. Find a subgame perfect Nash equilibrium and determine how long they wait.
Solution.
The last possible valuable proposal is at $t = 5$, made by Bernie. If Albert rejects at $t = 5$, both get 0. Hence Bernie can keep the whole remaining cake:

$$V_5^P = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Applying backward induction. At time t, the responder can reject and become proposer at time $t+1$. Therefore the responder must receive exactly the value with which the responder would accept the proposal

$$V_{t+1}^P -$$

Hence

$$V_t^P = \left(\frac{1}{2}\right)^t - V_{t+1}^P$$

Now compute with $V_6 = 0$:

$$V_5^P = \left(\frac{1}{2}\right)^5 - V_6 = \frac{1}{32} - 0 = \frac{1}{32},$$

$$V_4^P = \left(\frac{1}{2}\right)^4 - V_5 = \frac{1}{16} - \frac{1}{32} = \frac{1}{32},$$

$$V_3^P = \left(\frac{1}{2}\right)^3 - V_4 = \frac{1}{8} - \frac{1}{16} = \frac{3}{32},$$

$$V_2^P = \left(\frac{1}{2}\right)^2 - V_3 = \frac{1}{4} - \frac{3}{32} = \frac{5}{32},$$

$$V_1^P = \left(\frac{1}{2}\right)^1 - V_2 = \frac{1}{2} - \frac{5}{32} = \frac{11}{32},$$

$$V_0^P = \left(\frac{1}{2}\right)^0 - V_1 = \frac{21}{32}$$

At $t = 0$, Albert is proposer. He gives Bernie her continuation value:

$$\frac{11}{32}$$

Therefore Albert proposes

$$\left(\frac{21}{32}, \frac{11}{32}\right),$$

and Bernie accepts.
Thus the subgame perfect Nash equilibrium outcome is

$$\left(\frac{21}{32}, \frac{11}{32}\right)$$

and they wait

$$0 \text{ hours}$$

5. (Pirates and treasure). 50 pirates ordered by rank are dividing a big treasure that consists of 1000 equal indivisible items. They employ the following procedure. Firstly, the captain offers a division. If at least half of the crew (including the captain) agree, then the division is applied. Otherwise the captain is thrown overboard and the second-ranked pirate offers a division, and so on. Formalize the situation as a dynamic game and find a subgame perfect Nash equilibrium such that no pirate is indifferent at any time.
Solution.
Consider a subgame with n surviving pirates, relabeled

$$1, 2, \ldots, n,$$

where pirate 1 is the proposer. A proposal passes if it gets at least

$$\left\lceil \frac{n}{2} \right\rceil$$

votes. Since the proposer votes for himself, he needs

$$\left\lceil \frac{n}{2} \right\rceil - 1 = \left\lfloor \frac{n-1}{2} \right\rfloor$$

additional votes.
Using backward induction. A pirate votes yes only if he gets strictly more than he would get after rejecting the proposal. We also don't want to pay the next possible captain and we pay only to those pirates who would get 0 if a proposal is rejected.
For small n, the pattern is:

$$n = 1 : \quad (1000),$$

$$n = 2 : \quad (1000, 0),$$

$$n = 3 : \quad (999, 0, 1),$$

$$n = 4 : \quad (999, 0, 1, 0),$$

$$n = 5 : \quad (998, 0, 1, 0, 1)$$

Thus, in a subgame with n pirates, the proposer buys the cheapest votes: pirates

$$3, 5, 7, \ldots$$

each get 1 item, and the others get 0.
For $n = 50$, the captain needs

$$\left\lceil \frac{50}{2} \right\rceil = 25$$

votes total. His own vote counts, so he needs 24 more votes.

He gives 1 item to pirates
$$3, 5, 7, \dots, 49.$$
There are 24 such pirates. Therefore the captain keeps
$$1000 - 24 = 976.$$
Hence the equilibrium proposal is
$$(x_1, x_2, \dots, x_{50}) = (976, 0, 1, 0, 1, 0, \dots, 1, 0)$$
where
$$x_1 = 976,$$
$$x_3 = x_5 = \cdots = x_{49} = 1,$$
and
$$x_2 = x_4 = \cdots = x_{50} = 0$$
The yes-voters are
$$\{1, 3, 5, \dots, 49\},$$
which gives exactly 25 votes, so the proposal passes.
Therefore, in the subgame perfect Nash equilibrium, the first captain survives and keeps
$$976$$
items.